

# What Factors Influence Loan Approval

EXPLORATORY DATA ANALYSIS ON LOAN APPROVAL FACTORS

MAXIM BILENKIN

COURSE: DSC 530, PROF. FADI ALSALEEM

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# Introduction & Dataset Overview

## Introduction

### Understanding Loan Approval Factors

- ▶ Many individuals apply for loans, yet not all are approved.
- ▶ This project explores the key factors influencing loan approvals.

## Dataset Overview

- ▶ **Size:** 44,972 loan applications
- ▶ **Key Variables:** Age, Income, Loan Amount, Credit Score, etc.
- ▶ **Data Cleaning:** Removed extreme outliers (ages 109-144, incomes over \$1M).

# Descriptive Statistics

## Overview

- ▶ Descriptive statistics summarize key aspects of the dataset.
- ▶ They help understand the distribution, central tendency, and variability of the data.

## Key Statistics (After Data Cleaning)

- ▶ Total Records: 44,972 loan applications
- ▶ Mean Income: \$79,248
- ▶ Mean Loan Amount: \$9,582
- ▶ Credit Score Range: 390 – 784
- ▶ Loan Interest Rate Range: 5.42% – 20.00%

## Skewness Analysis

- ▶ Several variables, such as income and employment experience, exhibit skewness.
- ▶ Right-skewed distributions indicate the presence of extreme values affecting the mean.

## Next Steps

- ▶ Visualizing distributions to further explore trends in the dataset.

|      | age | income  | emp_exp | loan   | rate  | loan_to_inc | cred_hist | score |
|------|-----|---------|---------|--------|-------|-------------|-----------|-------|
| mean | 28  | 79,248  | 5       | 9,582  | 11.01 | 13.98       | 6         | 633   |
| std  | 6   | 54,684  | 6       | 6,313  | 2.98  | 8.72        | 4         | 50    |
| min  | 20  | 8,000   | 0       | 500    | 5.42  | 0.00        | 2         | 390   |
| 25%  | 24  | 47,172  | 1       | 5,000  | 8.59  | 7.00        | 3         | 601   |
| 50%  | 26  | 67,040  | 4       | 8,000  | 11.01 | 12.00       | 4         | 640   |
| 75%  | 30  | 95,698  | 8       | 12,236 | 13.00 | 19.00       | 8         | 670   |
| max  | 94  | 994,605 | 76      | 35,000 | 20.00 | 66.00       | 30        | 784   |

Skewness of Columns:

| Column      | Skewness |
|-------------|----------|
| age         | 1.91     |
| income      | 4.61     |
| emp_exp     | 1.95     |
| loan        | 1.18     |
| rate        | 0.21     |
| loan_to_inc | 1.04     |
| cred_hist   | 1.63     |
| score       | -0.62    |

# Distribution of Key Variables

## Understanding Data Distribution

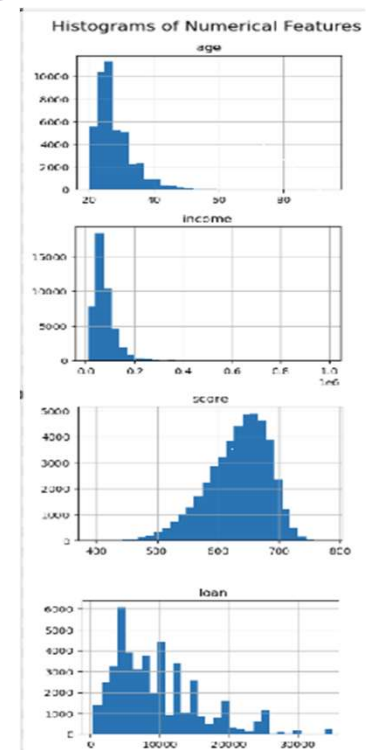
- ▶ Visualizing distributions helps identify patterns, skewness, and outliers.
- ▶ Key variables analyzed: Age, Income, Loan Amount, and Credit Score.

## Findings from Histograms & PMFs

- ▶ **Age:** Most applicants are between 24-35 years old.
- ▶ **Income:** Right-skewed distribution; a few individuals earn significantly higher than the median.
- ▶ **Loan Amount:** Most loans fall between \$5,000 - \$15,000, with fewer high-value loans.
- ▶ **Credit Score:** Normal-like distribution, peaking around 640-680.

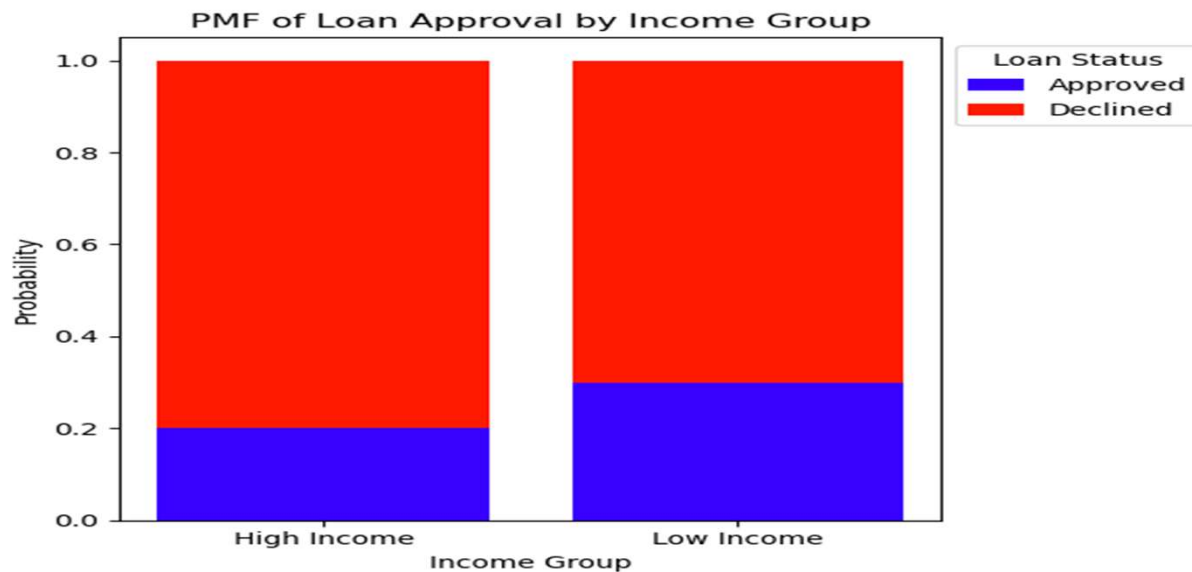
## Next Steps

- ▶ Use CDFs (Cumulative Distribution Functions) for deeper insights into variable distributions.



## PMF Analysis of Loan Approval by Income Group

- ▶ PMF shows the probability of discrete outcomes (approval/rejection) in different income groups.



- ▶ **Contrary to expectations**, low-income applicants have a higher loan approval rate than high-income applicants.

# CDF Analysis of Key Variables

## Why Use CDFs?

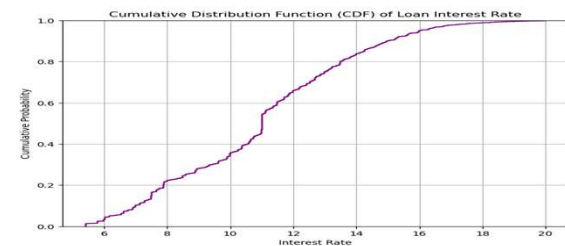
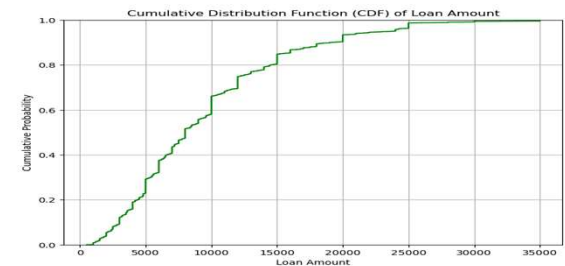
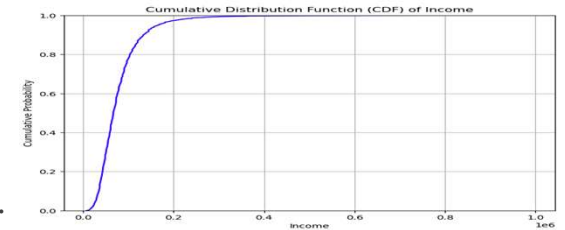
- ▶ CDFs help understand the proportion of data below a given threshold.
- ▶ They provide a clearer view of variable distribution compared to histograms.

## Findings from CDFs

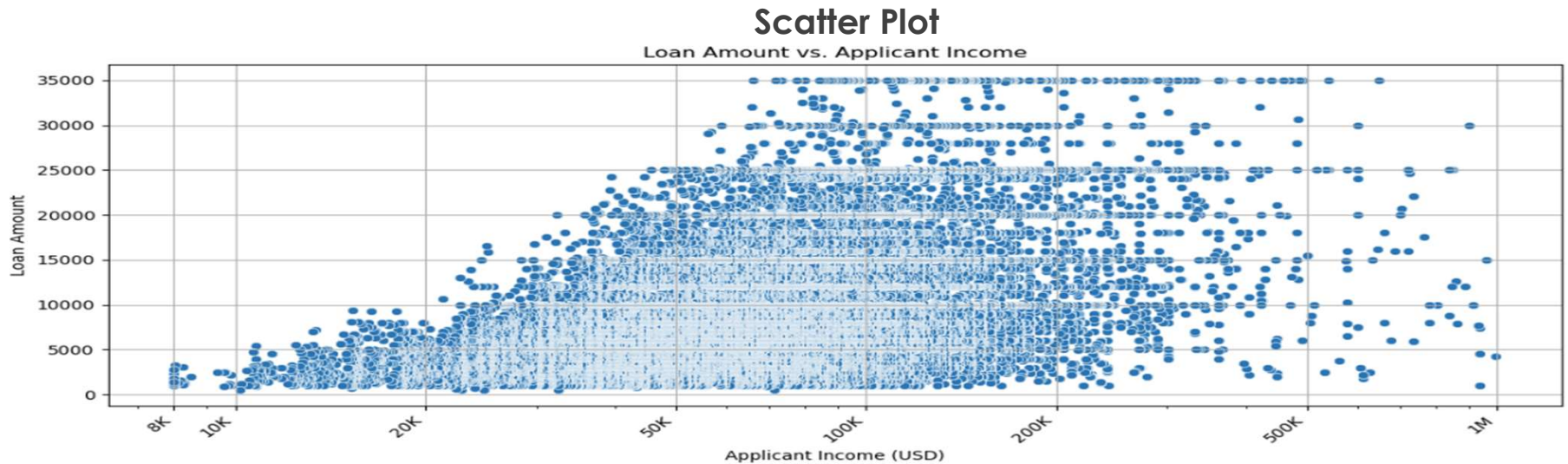
- ▶ **Age:** Around 50% of applicants are under 30 years old.
- ▶ **Income:** The majority earn below \$100,000, but a small percentage earns significantly more.
- ▶ **Loan Amount:** Most loans are under \$15,000, with a few high-value loans affecting the distribution.
- ▶ **Credit Score:** 75% of applicants have a credit score above 600.

## Next Steps

- ▶ Examine correlations between variables to identify relationships.



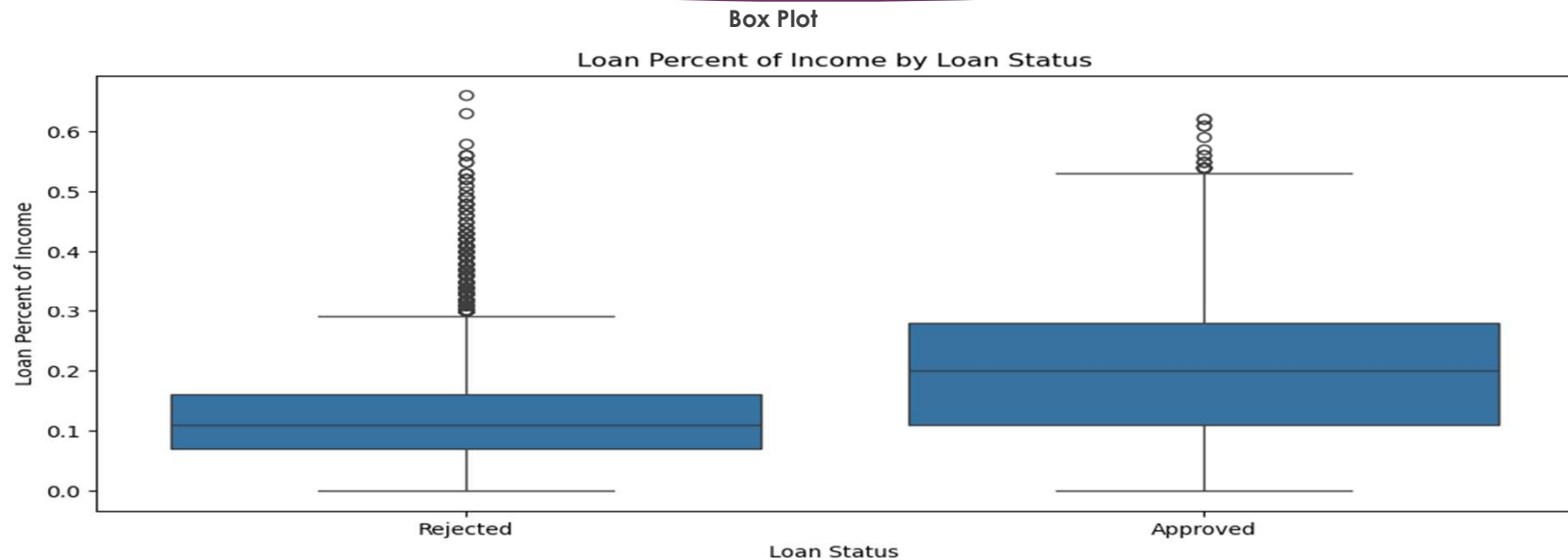
# Statistical Analysis & Predictions



## Brief Observations

- There is a moderate positive correlation ( $r = 0.36$ ) between loan amount and applicant income, meaning higher-income applicants tend to receive larger loans. This relationship is statistically significant.

# Statistical Analysis & Predictions



## Brief Observations

- There is a significant relationship between the loan amount as a percentage of income and loan status (**p-value = 0.0**). Higher loan percentages relative to income are associated with a higher likelihood of loan denial. Higher loan-to-income ratios show more variability among approved loans, but extremely high ratios increase the likelihood of rejection.



# Key Findings & Insights

## Loan Amount vs. Applicant Income

- ▶ **Income Correlation:** Higher-income applicants request and receive higher loan amounts.
- ▶ **Correlation Coefficient: 0.36** (Statistically significant,  $p < 0.0001$ ). A correlation of 0.36 indicates a moderate positive relationship - higher income tends to be associated with higher loan amounts, but other factors also play a role.
- ▶ **Key Insight:** Higher-income applicants are more likely to be approved for larger loans.

## Loan Percent of Income by Loan Status

- ▶ **Negative Relationship:** A higher loan amount as a percentage of annual income is more likely to be rejected.
- ▶ **T-statistic: 88.39** (Statistically significant,  $p < 0.0001$ ).
- ▶ **Key Insight:** The ratio of the loan amount to annual income is crucial when making approval decisions. Borrowers requesting loans that are a higher percentage of their income are less likely to be approved.

# Challenges & Limitations

## **Despite valuable insights, the analysis has some limitations**

- ▶ Most variables are positively skewed, which could impact statistical assumptions.
- ▶ Patterns in the data don't necessarily imply cause-and-effect relationships.
- ▶ Hypothesis testing and regression analysis have assumptions that may not hold perfectly.
- ▶ Economic conditions, policy changes, or applicant behavior could influence results.

# Conclusions & Implications

## Key Conclusions

### 1. Loan as a Percentage of Applicant Income

- ▶ **Strong Influence:** Loan-to-income ratio impacts approval decisions, but there is no strict cutoff rate.
- ▶ **Insight:** Loans with a higher percentage of income **show an increasing rejection trend beyond 20-30%**, but approval rates improve for applicants beyond 30-40%, suggesting other factors also play a role.

### 2. Loan Amount vs. Applicant Income

- ▶ **Positive Correlation:** A clear relationship exists between applicant income and loan amount.
- ▶ **Insight:** Higher-income applicants tend to receive higher loan amounts.

## Implications

- ▶ **For Lenders:** These key factors can help refine the loan approval process and improve decision-making.
- ▶ **For Borrowers:** Maintaining a balanced loan-to-income ratio is beneficial, but other creditworthiness factors, such as credit score and lender policies, also affect approval.